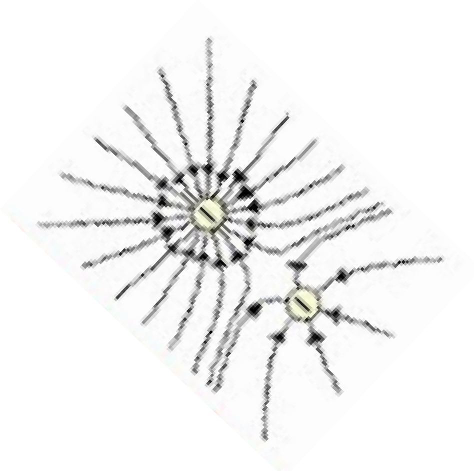
9	(a)		The principle of moments states that for a body to be in (rotational) equilibrium, the sum of clockwise moments about any point must be equal to the sum of anti-clockwise moments about the same point. (Accept net/total moment about a fixed point is zero)	
	(b)	(i)	As sphere X is heavier than mass A, the clockwise moment due to the weight of sphere X about P is greater than the anticlockwise due to the weight of mass A about P. By the principle of moments, for mass A and sphere X to be balanced, the electric force on sphere X must provide an anti-clockwise moment. Sphere Y must be negatively charged to provide such an electric force on sphere X.	
		(ii)	Let the electric force be F and l the length of the rod. By the principle of moments, Clockwise moments about P = anti-clockwise moments about P $0.200 \times 9.81 \times \frac{l}{2} = 0.150 \times 9.81 \times \frac{l}{2} + Fsin25° \times \frac{l}{2}$ $Fsin25° = 0.4905\ N$ $F = 1.16N$	
		(iii)	Pointing upwards and to the right	

	(c)			
	(d)	(i)	The potential energy of the decreases as it is moved from d = 0.05 m and reaches a minimum potential energy in between d = 0.05 m to d = 0.35 m. It then increases as it is moved towards d = 0.35 m.	
		(ii)	Potential due to X = - 150kV Potential due to Y = - 370 kV Total potential = - 150 – 370 = - 520 kV Potential energy = qV = $-520000 \times (-1.6 \times 10^{-19}) = 8.32 \times 10^{-14} J$	
		(iii)	Negative potential gradient is the electric field strength Since net force is zero, the net electric field strength is zero. Hence, the magnitude of the potential gradient of X is equal to that of Y at this point The direction of the potential gradient of X is opposite to that of Y at this point.	